Optimization In AI

Class 03: Gradient Descent

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Announcements

1. Class Rescheduling

The **November 4** class will be rescheduled.

2. Homework 1

Released **today**.

Two weeks to complete.



Rescheduling Survey

Today's Topics

01 Descent And Stationarity

02 Smooth Convex Functions

03 Strong Convexity

04 Nonsmooth Convex Functions

05 Frontier: Large Steps

Convergence Rates And Error Metrics

Assumptions / Method	Step Size	Certified Quantity And Rate
Smooth, lower bounded	$1/L$	$\min_{0 \le t < T} \|\nabla f(x_t)\|^2 = O(1/T)$
Smooth convex, minimizer	$1/L$	Last gap: $f(x_T) - f^\star = O(1/T)$
Smooth strongly convex	$1/L$	Last gap: $O((1 - \mu/L)^T)$
Convex subgradient, fixed	$R/(G\sqrt{T})$	Averaged gap $\le RG/\sqrt{T}$

PART 01

Descent And Stationarity

Gradient Descent From A Quadratic Upper Bound

$f : \mathbb{R}^d \to \mathbb{R}$ has an L-Lipschitz gradient, $L > 0$.

Lemma (Descent Lemma)

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2.$$

Minimize The Model In y

$$0 = \nabla f(x) + L(y - x) \quad \Longrightarrow \quad \boxed{x^+ = x - \frac{1}{L}\nabla f(x).}$$

Gradient Descent: One-Step Decrease

$x^+ = x - \eta \nabla f(x)$

$$f(x^+) \leq f(x) - \underbrace{\eta \left(1 - \frac{L\eta}{2}\right)}_{c(\eta)} \|\nabla f(x)\|^2.$$

Maximize $c(\eta)$ $\quad \eta \geq 0$

$$c'(\eta) = 1 - L\eta = 0 \quad \Longrightarrow \quad \eta = 1/L.$$

Lemma (One-Step Decrease Of Gradient Descent)

$$f(x^+) \leq f(x) - \frac{1}{2L}\|\nabla f(x)\|^2.$$

Nonconvex GD: Stationarity Rate

Theorem (Gradient Descent: Stationarity Bound)

Suppose f is L-smooth and $f^\star := \inf_{x\in\mathbb{R}^d} f(x) > -\infty$. For $x_{t+1} = x_t - \nabla f(x_t)/L$ and every integer $T \geq 1$,

$$\min_{0\leq t<T} \|\nabla f(x_t)\|^2 \leq \frac{1}{T}\sum_{t=0}^{T-1} \|\nabla f(x_t)\|^2 \leq \frac{2L(f(x_0)-f^\star)}{T}.$$

$$\min_{t<T} \|\nabla f(x_t)\| = O(T^{-1/2}) \quad \Longrightarrow \quad T = O(\epsilon^{-2}) \quad \text{for } \epsilon\text{-stationarity.}$$

Proof: Bounding The Sum Of Squared Gradients

One-Step Progress $\eta = 1/L$

$$\frac{1}{2L}\|\nabla f(x_t)\|^2 \leq f(x_t) - f(x_{t+1}).$$

Telescope

$$\begin{aligned}\frac{1}{2L}\sum_{t=0}^{T-1}\|\nabla f(x_t)\|^2 &\leq f(x_0) - f(x_T) \\ &\leq f(x_0) - f^\star.\end{aligned}$$

Minimum $\leq$ Average

$$\min_{0\leq t<T}\|\nabla f(x_t)\|^2 \leq \frac{2L(f(x_0) - f^\star)}{T}.$$

PART 02

Smooth Convex Functions

Smooth Convex GD: Convergence Rate

Theorem (Gradient Descent: Sublinear Convergence)

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and L-smooth, $L > 0$, with minimizer $x^\star$. For $x_{t+1} = x_t - \nabla f(x_t)/L$, let $R = \|x_0 - x^\star\|$. For every integer $T \geq 1$,

$$f(x_T) - f^\star \leq \frac{2LR^2}{T}.$$

Proof: From Descent To A Gap Recurrence

$R > 0, \qquad \Delta_t = f(x_t) - f^\star$

Smoothness

$$\Delta_{t+1} \le \Delta_t - \frac{1}{2L}\|\nabla f(x_t)\|^2.$$

Convexity

$\|x_t - x^\star\| \le R$

$$\Delta_t \le \langle \nabla f(x_t), x_t - x^\star \rangle \le \|\nabla f(x_t)\| \, \|x_t - x^\star\| \le R\|\nabla f(x_t)\|.$$

$$\boxed{\Delta_{t+1} \le \Delta_t - \frac{\Delta_t^2}{2LR^2}.}$$

Proof: Distance To A Minimizer Does Not Increase

Lemma (Nonincreasing Distance To A Minimizer)

For every $t \geq 0$, $\|x_{t+1} - x^\star\| \leq \|x_t - x^\star\| \leq R$.

Expand The Update

$$\begin{aligned}\|x_{t+1} - x^\star\|^2 &= \|x_t - x^\star\|^2 - \frac{2}{L}\langle \nabla f(x_t), x_t - x^\star\rangle + \frac{\|\nabla f(x_t)\|^2}{L^2} \\ &\leq \|x_t - x^\star\|^2 - \frac{2\Delta_t}{L} + \frac{2\Delta_t}{L} \\ &= \|x_t - x^\star\|^2.\end{aligned}$$

Inequality: Convexity + Descent

$$\langle \nabla f(x_t), x_t - x^\star\rangle \geq \Delta_t, \qquad \frac{\|\nabla f(x_t)\|^2}{2L} \leq \Delta_t - \Delta_{t+1} \leq \Delta_t.$$

Induction: $\|x_t - x^\star\| \leq R$.

Proof: Reciprocal Gaps Give The $1/T$ Rate

If $R = 0$ or $\Delta_T = 0$, the claim is immediate. Assume $\Delta_T > 0$; $C = 2LR^2 > 0$.

Smoothness + Descent

$$0 < \Delta_t \le \Delta_0 \le \tfrac{L}{2}R^2 = \tfrac{C}{4}.$$

Reciprocals $1 - \Delta_t/C > 0$

$$\frac{1}{\Delta_{t+1}} \ge \frac{1}{\Delta_t(1-\Delta_t/C)} \ge \frac{1}{\Delta_t} + \frac{1}{C}, \qquad \frac{1}{1-z} \ge 1 + z \quad (0 \le z < 1).$$

Telescope

$$\frac{1}{\Delta_T} \ge \frac{1}{\Delta_0} + \frac{T}{2LR^2} \quad \Longrightarrow \quad \boxed{\Delta_T \le \frac{1}{\Delta_0^{-1} + T/(2LR^2)} \le \frac{2LR^2}{T}.}$$

Potential Proof: The Last-Iterate Gap Bound

Definition (Smooth Convex Potential)

$$\Phi_t = t\Delta_t + \frac{L}{2}D_t^2, \qquad D_t = \|x_t - x^\star\|.$$

If $\Phi_{t+1} \le \Phi_t$ For $0 \le t < T$

$$\Phi_T \le \Phi_{T-1} \le \cdots \le \Phi_0 = \frac{L}{2}R^2.$$

$$T\Delta_T \le T\Delta_T + \frac{L}{2}D_T^2 = \Phi_T \le \frac{L}{2}R^2.$$

Corollary (Gradient Descent: Last-Iterate Bound)

$$\Delta_T \le \frac{LR^2}{2T} \qquad (T \ge 1).$$

Potential Proof: Monotonicity For Convex GD

Distance Change $\eta = 1/L$

$$\begin{aligned}\frac{L}{2}(D_t^2 - D_{t+1}^2) &= \langle \nabla f(x_t), x_t - x^\star \rangle - \frac{\|\nabla f(x_t)\|^2}{2L} \\ &\geq \Delta_t - \frac{\|\nabla f(x_t)\|^2}{2L} && \text{convexity} \\ &\geq \Delta_{t+1} && \text{descent.}\end{aligned}$$

Potential Change

$$\begin{aligned}\Phi_{t+1} - \Phi_t &= t(\Delta_{t+1} - \Delta_t) + \Delta_{t+1} + \frac{L}{2}(D_{t+1}^2 - D_t^2) \\ &\leq t(\Delta_{t+1} - \Delta_t) \leq 0 \qquad \text{by descent.}\end{aligned}$$

Example: Linear Regression

$A \in \mathbb{R}^{n \times d}, \quad b \in \mathbb{R}^n$

$$f(x) = \frac{1}{2}\|Ax - b\|^2.$$

Gradient + Curvature

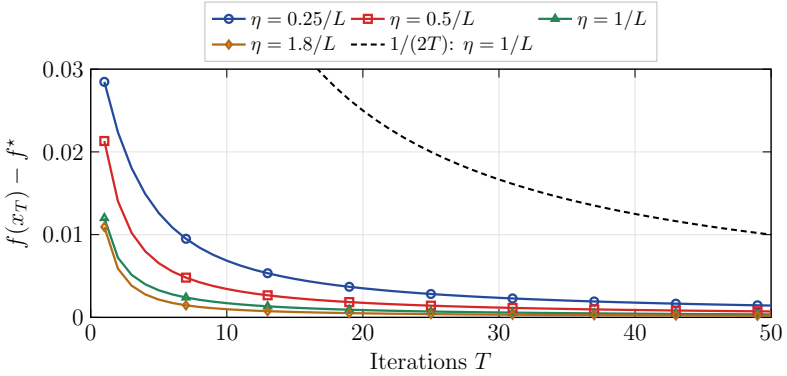
$$\nabla f(x) = A^\top (Ax - b), \qquad \nabla^2 f(x) = A^\top A \succeq 0.$$

Smooth Convex Structure

$L = \|A\|_2^2$ $(A \neq 0)$, $\qquad \arg\min f \neq \varnothing$

Linear Regression: Objective Gap And Stepsize

$$d = 200, \quad A = \mathrm{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_d}), \quad b = 0$$
$$\lambda_j = 10^{-6(j-1)/(d-1)} \ (j = 1, \ldots, d), \quad x_0 = \mathbf{1}/\sqrt{d}, \quad L = 1$$



PART 03

Smooth Strongly Convex Functions

Strongly Convex GD: Geometric Convergence

Theorem (Gradient Descent: Linear Convergence)

Let $f : \mathbb{R}^d \to \mathbb{R}$ be L-smooth and μ-strongly convex, $0 < \mu \leq L$, with minimizer $x^\star$. GD with $\eta = 1/L$ satisfies, for every integer $T \geq 1$,

$$f(x_T) - f^\star \leq \left(1 - \frac{\mu}{L}\right)^T (f(x_0) - f^\star).$$

Proof: Strong Convexity Gives Gap Contraction

Descent $\eta = 1/L$

$$\Delta_{t+1} \le \Delta_t - \frac{1}{2L}\|\nabla f(x_t)\|^2$$
$$\le \Delta_t - \frac{\mu}{L}\Delta_t = \left(1 - \frac{\mu}{L}\right)\Delta_t, \qquad \|\nabla f(x_t)\|^2 \ge 2\mu\Delta_t.$$

$\kappa = L/\mu$

$$\boxed{\Delta_T \le (1 - \mu/L)^T \Delta_0} \le e^{-T/\kappa}\Delta_0.$$

Geometry	Guaranteed Decrease	Gap Rate
Convex, $R > 0$	$\Delta_t^2/(2LR^2)$	$O(1/T)$
Strongly convex	$(\mu/L)\Delta_t$	Geometric

Proof: Bounding The Gap By The Squared Gradient

Strong Convexity At $x^\star$

$$\begin{aligned}\Delta_t &\le \langle \nabla f(x_t), x_t - x^\star \rangle - \frac{\mu}{2} D_t^2 && \text{strong convexity} \\ &\le \|\nabla f(x_t)\| D_t - \frac{\mu}{2} D_t^2 && \text{Cauchy–Schwarz.}\end{aligned}$$

Complete The Square In D_t

$$\|\nabla f(x_t)\| D_t - \frac{\mu}{2} D_t^2 = \frac{\|\nabla f(x_t)\|^2}{2\mu} - \frac{\mu}{2}\left(D_t - \frac{\|\nabla f(x_t)\|}{\mu}\right)^2 \le \frac{\|\nabla f(x_t)\|^2}{2\mu}.$$

$$\boxed{\|\nabla f(x_t)\|^2 \ge 2\mu \Delta_t.}$$

Strongly Convex GD: An Exponential Potential

$0 < \mu < L, \qquad q = 1 - \mu/L \in (0, 1)$

Definition (Strongly Convex Potential)

$$\Phi_t = q^{-t}\Delta_t.$$

If $\Phi_{t+1} \le \Phi_t$ For $0 \le t < T$

$$\Phi_T \le \cdots \le \Phi_0 = \Delta_0 \quad \Longrightarrow \quad \boxed{\Delta_T = q^T \Phi_T \le q^T \Delta_0.}$$

Monotonicity: $\Delta_{t+1} \le q\Delta_t$

$$\begin{aligned}\Phi_{t+1} - \Phi_t &= q^{-(t+1)}\Delta_{t+1} - q^{-t}\Delta_t \\ &= q^{-(t+1)}(\Delta_{t+1} - q\Delta_t) \le 0.\end{aligned}$$

$\mu = L$: $\qquad 0 \le \Delta_{t+1} \le 0 \quad \Longrightarrow \quad \Delta_T = 0 \ (T \ge 1).$

Strongly Convex GD: Distance Convergence

Theorem (Gradient Descent: Distance Contraction)

Let $f \in C^2(\mathbb{R}^d)$ be μ-strongly convex and L-smooth, $0 < \mu \le L$, with minimizer $x^\star$. For GD with constant $0 < \eta < 2/L$,

$$r_\eta := \max\{|1 - \eta\mu|,\ |1 - \eta L|\} < 1,$$

$$\boxed{\|x_T - x^\star\| \le r_\eta^T \|x_0 - x^\star\|}, \qquad T \ge 1.$$

Step Size	Distance Factor r_η
$1/L$	$1 - \mu/L$
$2/(L+\mu)$	$(L-\mu)/(L+\mu)$

Proof: Average The Hessian Along A Segment

$e_t = x_t - x^\star, \qquad z_t(s) = x^\star + s e_t, \quad 0 \le s \le 1$

Fundamental Theorem Of Calculus $\quad \nabla f(x^\star) = 0$

$$\begin{aligned}\nabla f(x_t) &= \nabla f(x_t) - \nabla f(x^\star) \\ &= \int_0^1 \nabla^2 f(z_t(s))\, e_t \, ds = \overline{H}_t e_t, \qquad \overline{H}_t := \int_0^1 \nabla^2 f(z_t(s))\, ds.\end{aligned}$$

Strong Convexity + Smoothness $\quad \forall x$

$$\mu I \preceq \nabla^2 f(x) \preceq L I.$$

Integrate The Bounds

$$\boxed{\mu I \preceq \overline{H}_t \preceq L I}, \qquad \overline{H}_t = \overline{H}_t^\top.$$

Proof: Contract The Distance By A Spectral Bound

Exact Error Update

$$e_{t+1} = e_t - \eta \nabla f(x_t) = (I - \eta \overline{H}_t) e_t.$$

Symmetry + Eigenvalues In $[\mu, L]$

$$\begin{aligned} \|e_{t+1}\| &\le \|I - \eta \overline{H}_t\|_2 \, \|e_t\| \\ &= \max_i |1 - \eta \lambda_i(\overline{H}_t)| \, \|e_t\| \\ &\le \underbrace{\max\{|1 - \eta\mu|, |1 - \eta L|\}}_{r_\eta} \, \|e_t\|. \end{aligned}$$

Iteration

$$\boxed{\|x_T - x^\star\| \le r_\eta^T \|x_0 - x^\star\|}.$$

$$\eta = \tfrac{2}{L+\mu} : \; r_\eta = \tfrac{L-\mu}{L+\mu}. \qquad \mu = L : \; \eta = 1/L \;\Rightarrow\; x_1 = x^\star.$$

Example: Ridge Regression

Quadratic Regularization

$$f_\lambda(x) = \frac{1}{2}\|Ax - b\|^2 + \frac{\lambda}{2}\|x\|^2, \qquad \lambda > 0.$$

Update

$$x_{t+1} = x_t - \eta \left[A^\top (Ax_t - b) + \lambda x_t \right].$$

Curvature

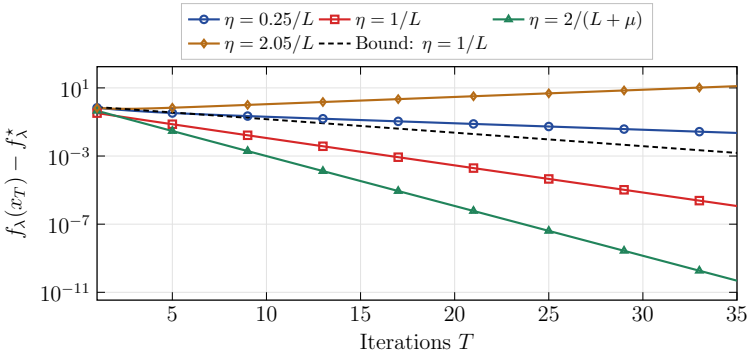
$$L = \lambda_{\max}(A^\top A) + \lambda, \qquad \mu = \lambda_{\min}(A^\top A) + \lambda > 0.$$

$$\eta = 1/L \qquad \Rightarrow \qquad q = 1 - \mu/L$$

Ridge Regression: Objective Gap And Stepsize

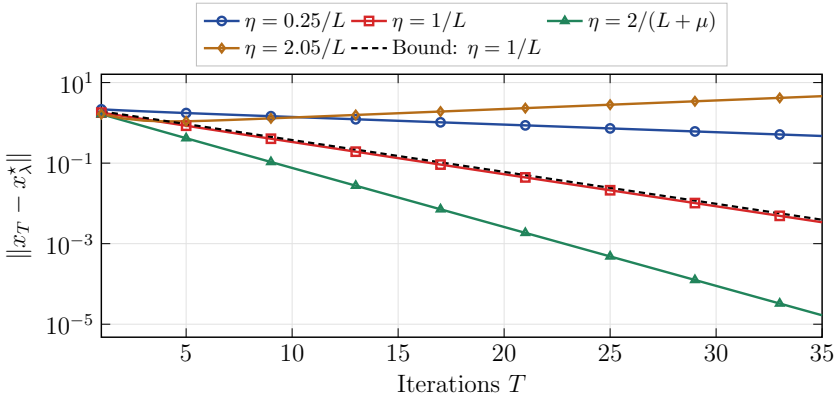
$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/5 & 0 \end{pmatrix}, \quad b = (1,1)^\top, \quad x_0 = (0,0,2)^\top$$

$$\lambda = 1/5, \quad L = 6/5, \quad \mu = 1/5, \quad x_\lambda^\star = (5/6, 5/6, 0)^\top$$



Ridge Regression: Distance To The Minimizer

$$\lambda = 1/5, \quad L = 6/5, \quad \mu = 1/5, \quad x_0 = (0, 0, 2)^\top$$



PART 04

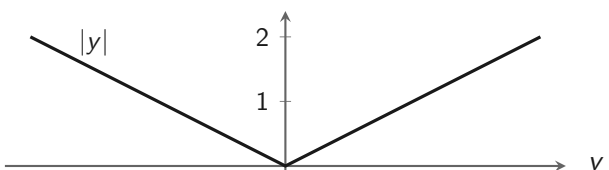
Nonsmooth Convex Functions

Convex Subgradients: Supporting Affine Bounds

Definition (Convex Subgradient)

For convex $f : \mathbb{R}^d \to \mathbb{R}$,

$$g \in \partial f(x) \iff f(y) \geq f(x) + \langle g, y - x \rangle \quad \forall y \in \mathbb{R}^d.$$



Subgradient Method

Setting

Minimize a convex function $f : \mathbb{R}^d \to \mathbb{R}$; start from $x_0 \in \mathbb{R}^d$.

Algorithm

For a constant stepsize $\eta > 0$,

$$\boxed{x_{t+1} = x_t - \eta g_t, \qquad g_t \in \partial f(x_t).}$$

Output After T Steps

$$\bar{x}_T = \frac{1}{T} \sum_{t=0}^{T-1} x_t.$$

Subgradient Method: Averaged-Iterate Rate

Assumptions

$f : \mathbb{R}^d \to \mathbb{R}$ is convex and has a minimizer $x^\star$.

$$\|x_0 - x^\star\| \le R, \qquad g_t \in \partial f(x_t), \quad \|g_t\| \le G \quad (0 \le t < T).$$

Theorem (Subgradient Method: Averaged-Iterate Bound)

For integer $T \ge 1$ and constant $\eta > 0$, use $x_{t+1} = x_t - \eta g_t$ and return the uniform average $\bar{x}_T$. Then

$$f(\bar{x}_T) - f^\star \le \frac{R^2}{2\eta T} + \frac{\eta G^2}{2}.$$

$$R, G > 0 : \quad \eta = \frac{R}{G\sqrt{T}} \quad \Longrightarrow \quad f(\bar{x}_T) - f^\star \le \frac{RG}{\sqrt{T}}.$$

Proof: One-Step Distance Bound For Subgradients

1. Expand The Squared Distance

$$D_{t+1}^2 = D_t^2 - 2\eta\langle g_t, x_t - x^\star\rangle + \eta^2\|g_t\|^2.$$

2. Apply The Subgradient Inequality

$$f^\star \geq f(x_t) + \langle g_t, x^\star - x_t\rangle \quad \Longrightarrow \quad \langle g_t, x_t - x^\star\rangle \geq \Delta_t.$$

Using $\|g_t\| \leq G$,

$$D_{t+1}^2 \leq D_t^2 - 2\eta\Delta_t + \eta^2 G^2.$$

$$\boxed{2\eta\Delta_t \leq D_t^2 - D_{t+1}^2 + \eta^2 G^2.}$$

Proof: Telescoping Gives The Averaged Gap Bound

3. Telescope The Distance Terms

$$2\eta \sum_{t=0}^{T-1} \Delta_t \le D_0^2 - D_T^2 + T\eta^2 G^2 \le R^2 + T\eta^2 G^2.$$

Jensen's Inequality

$$f(\bar{x}_T) - f^\star \le \frac{1}{T} \sum_{t=0}^{T-1} \Delta_t \le \frac{R^2}{2\eta T} + \frac{\eta G^2}{2}.$$

Minimizing The Bound $R, G > 0$

$$\eta = \frac{R}{G\sqrt{T}} \quad \Longrightarrow \quad \frac{R^2}{2\eta T} = \frac{\eta G^2}{2} = \frac{RG}{2\sqrt{T}}.$$

Diminishing Stepsizes: Weighted Averaging

$f : \mathbb{R}^d \to \mathbb{R}$ convex; $x^\star \in \arg\min f$; $\|x_0 - x^\star\| \le R$; $R, G > 0$.

$x_{t+1} = x_t - \eta_t g_t$, $\eta_t > 0$; along this run, $g_t \in \partial f(x_t)$, $\|g_t\| \le G$ for all $t \ge 0$.

Proof: Sum The Distance Bounds $T \ge 1$

$$2\eta_t \Delta_t \le D_t^2 - D_{t+1}^2 + \eta_t^2 G^2,$$

$$2\sum_{t=0}^{T-1} \eta_t \Delta_t \le R^2 + G^2 \sum_{t=0}^{T-1} \eta_t^2.$$

Jensen's Inequality

$$\boxed{\widehat{x}_T = \frac{\sum_{t=0}^{T-1} \eta_t x_t}{\sum_{t=0}^{T-1} \eta_t}.}$$

$$\boxed{f(\widehat{x}_T) - f^\star \le \frac{R^2 + G^2 \sum_{t=0}^{T-1} \eta_t^2}{2\sum_{t=0}^{T-1} \eta_t}.}$$

Subgradient Rate With Diminishing Steps

Set $\eta_t = R/(G\sqrt{t+1})$ $\quad t \geq 0$; every integer $T \geq 1$.
Use $\sum_{k=1}^{T} 1/k \leq 1 + \log T$ and $\sum_{k=1}^{T} 1/\sqrt{k} \geq \sqrt{T}$.

Corollary

$$f(\widehat{x}_T) - f^\star \leq \frac{RG\left(1 + \sum_{k=1}^{T} 1/k\right)}{2\sum_{k=1}^{T} 1/\sqrt{k}} \leq \frac{RG(2 + \log T)}{2\sqrt{T}}.$$

	Fixed Step	**Diminishing Step**
Learning rate	$R/(G\sqrt{T})$	$R/(G\sqrt{t+1})$
Needs horizon T?	Yes	No
Output	Uniform average $\bar{x}_T$	Stepsize-weighted $\widehat{x}_T$
Gap guarantee	$RG/\sqrt{T}$	$O(RG \log T/\sqrt{T})$

Subgradient Potential: Bounding Accumulated Gaps

Constant $\eta > 0$ fixed $T \geq 1$, $0 \leq t \leq T$.

Use the subgradient theorem's assumptions along this run.

Definition (Subgradient Potential)

$$\Phi_t = D_t^2 + 2\eta \sum_{s=0}^{t-1} \Delta_s + (T-t)\eta^2 G^2.$$

If $\Phi_{t+1} \leq \Phi_t$ For $0 \leq t < T$ $\Phi_T \leq \cdots \leq \Phi_0$.

$$\Phi_0 = D_0^2 + T\eta^2 G^2, \qquad \Phi_T = D_T^2 + 2\eta \sum_{t=0}^{T-1} \Delta_t.$$

$$2\eta \sum_{t=0}^{T-1} \Delta_t \leq \Phi_T \leq \Phi_0 \leq R^2 + T\eta^2 G^2.$$

Subgradient Potential: Rate And Monotonicity

Rate: Jensen's Inequality

$$f(\bar{x}_T) - f^\star \le \frac{1}{T}\sum_{t=0}^{T-1} \Delta_t \le \frac{R^2}{2\eta T} + \frac{\eta G^2}{2}.$$

Substitute $\eta = R/(G\sqrt{T})$ $\quad R, G > 0$

$$\boxed{f(\bar{x}_T) - f^\star \le \frac{RG}{2\sqrt{T}} + \frac{RG}{2\sqrt{T}} = \frac{RG}{\sqrt{T}}.}$$

Monotonicity: The Distance Bound

$$\begin{aligned}\Phi_{t+1} - \Phi_t &= D_{t+1}^2 - D_t^2 + 2\eta\Delta_t - \eta^2 G^2 \\ &\le \left(-2\eta\Delta_t + \eta^2\|g_t\|^2\right) + 2\eta\Delta_t - \eta^2 G^2 \\ &= \eta^2(\|g_t\|^2 - G^2) \le 0 \qquad (0 \le t < T).\end{aligned}$$

Example: Lasso

$$F_\lambda(x) = \frac{1}{2}\|Ax - b\|^2 + \lambda\|x\|_1, \qquad \lambda > 0.$$

Ordinary Subgradient Update

$$g_t = A^\top(Ax_t - b) + \lambda s_t, \quad s_t \in \partial\|x_t\|_1, \qquad x_{t+1} = x_t - \eta g_t.$$

A Sparse Optimizer: $A = I$

$b = (2, 0.3)^\top, \quad \lambda = 0.5$

$$x_j^\star = \operatorname{sign}(b_j)\max\{|b_j| - \lambda, 0\} \quad \Longrightarrow \quad x^\star = (1.5, 0)^\top.$$

Lasso: Bounded Iterates

Assumptions $A = I, \quad \lambda > 0, \quad 0 < \eta \leq 1, \quad s_t \in \partial \|x_t\|_1.$

Each coordinate of s_t lies in $[-1, 1]$, so $\|s_t\| \leq \sqrt{d}$.

One-Step Bound

$$x_{t+1} = (1 - \eta)x_t + \eta(b - \lambda s_t).$$

With $M = \|b\| + \lambda\sqrt{d}$,

$$\|x_{t+1}\| \leq (1 - \eta)\|x_t\| + \eta M.$$

Induction: Set $B = \max\{\|x_0\|, M\}$

Base: $\|x_0\| \leq B$. If $\|x_t\| \leq B$, then

$$\|x_{t+1}\| \leq (1 - \eta)B + \eta M \leq B.$$

$$\boxed{\|x_t\| \leq B \quad \text{for all } t \geq 0.}$$

Lasso: Bounded Subgradients

Same Setting $A = I, \quad 0 < \eta \leq 1.$

Subgradient Bound

$$\|g_t\| = \|x_t - b + \lambda s_t\| \leq B + \|b\| + \lambda\sqrt{d} =: G \quad \forall t \geq 0.$$

Admissible Step: $x_0 = 0$

Soft-thresholding gives $|x_j^\star| \leq |b_j|$, hence

$$R = \|x_0 - x^\star\| = \|x^\star\| \leq \|b\| \leq G.$$

For $R > 0$ and every integer $T \geq 1$, $0 < \eta = R/(G\sqrt{T}) \leq 1$, so

$$\boxed{F_\lambda(\bar{x}_T) - F_\lambda^\star \leq \frac{RG}{\sqrt{T}}.}$$

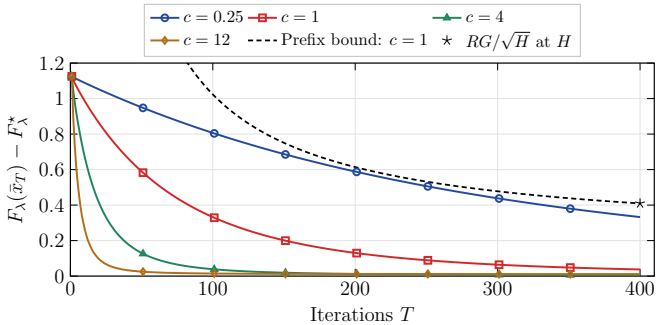
If $R = 0$, return the already optimal x_0.

Lasso: Subgradient Steps And Averaged Gap

$$A = I,\ b = (2, 0.3)^\top,\ \lambda = 0.5,\ x_0 = 0,\ H = 400$$

$$R = \|x^\star\|, \quad G = 2(\|b\| + \lambda\sqrt{2}), \quad \eta = cR/(G\sqrt{H}) \le 1 \quad \text{for all four runs}$$

$$s_t = \operatorname{sign}(x_t), \quad \operatorname{sign}(0) = 0$$

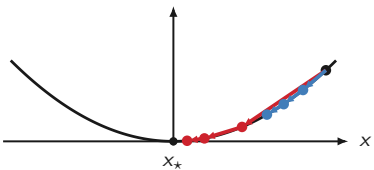


PART 05

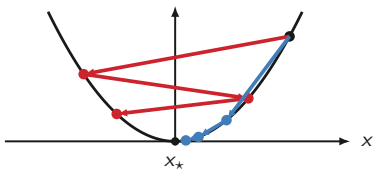
Stepsize-Based Acceleration

Stepsizes In Flat And Steep Directions

Flat direction



Steep direction



Large Steps Small Steps

Chebyshev Steps: Endpoint Convergence

Theorem (Euclidean Error In Exact Arithmetic)

Let $f(x) = f^\star + \frac{1}{2}(x - x^\star)^\top Q(x - x^\star)$, where

$$Q = Q^\top, \qquad 0 < \mu < L, \qquad \mu I \preceq Q \preceq LI.$$

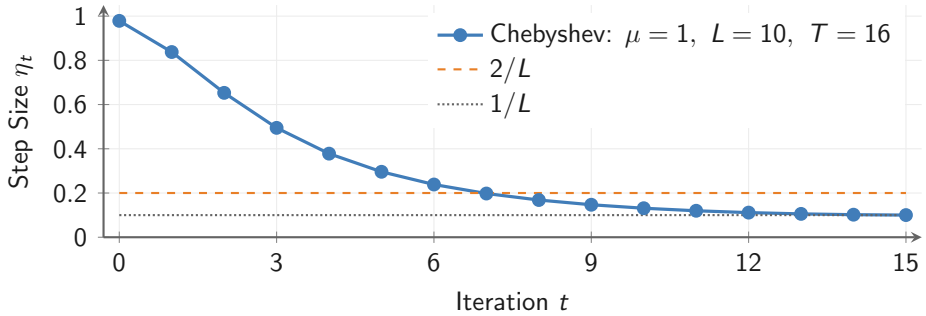
For every integer $T \geq 1$, there are explicit positive steps $\eta_0, \ldots, \eta_{T-1}$ depending only on μ, L, T such that, for every x_0, GD satisfies

$$\boxed{\|x_T - x^\star\| \leq 2\left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^T \|x_0 - x^\star\|, \qquad \kappa = \frac{L}{\mu}.}$$

Best constant-step GD	Chebyshev steps
$O(\kappa \log(1/\epsilon))$	$O(\sqrt{\kappa} \log(1/\epsilon))$

Chebyshev Stepsizes

$$\eta_t = \frac{2}{L + \mu - (L - \mu)\cos\left(\frac{(2t+1)\pi}{2T}\right)}, \qquad t = 0, \ldots, T-1, \quad 0 < \mu < L.$$



Quadratic GD As A Spectral Polynomial

Let $e_t = x_t - x^\star$. Since $\nabla f(x_t) = Qe_t$,

$$e_{t+1} = (I - \eta_t Q)e_t, \qquad e_T = p_T(Q)e_0, \qquad p_T(\lambda) = \prod_{t=0}^{T-1}(1 - \eta_t \lambda).$$

Spectral Theorem

$$\|e_T\| \le \max_{\lambda \in [\mu, L]} |p_T(\lambda)| \, \|e_0\|.$$

Interval Minimax Problem

$$\rho_T^\star = \min_{\substack{p \in \mathbb{R}[\lambda] \\ \deg(p) \le T,\ p(0)=1}} \max_{\lambda \in [\mu, L]} |p(\lambda)|.$$

Maximizing A Bounded Polynomial At c

Affine Change Of Variable

$$z = \frac{L+\mu-2\lambda}{L-\mu}, \qquad c = \frac{L+\mu}{L-\mu} > 1, \qquad [\mu, L] \mapsto [-1,1], \quad 0 \mapsto c.$$

Write $p(\lambda) = r(z)$ and $\|r\|_\infty = \max_{z\in[-1,1]} |r(z)|$. Then

$$\rho_T^\star = \min_{\substack{\deg(r)\le T \\ r(c)=1}} \|r\|_\infty.$$

Equivalent Maximization

$$\boxed{M_T = \max_{\substack{\deg(s)\le T \\ |s(z)|\le 1 \ \forall z\in[-1,1]}} s(c), \qquad \rho_T^\star = \frac{1}{M_T}.}$$

Indeed, $s = r/\|r\|_\infty$ gives $s(c) = 1/\|r\|_\infty$; conversely,
$r = s/s(c)$ gives $r(c) = 1$ and $\|r\|_\infty \le 1/s(c)$ when $s(c) > 0$.

Chebyshev Polynomials And Alternating Extrema

Definition: Chebyshev Polynomials

$$C_T(\cos\theta) = \cos(T\theta), \qquad |C_T(z)| \le 1 \quad (z \in [-1,1]).$$

Recurrence: Cosine Addition

$$C_0(z) = 1, \quad C_1(z) = z, \quad C_{m+1}(z) = 2zC_m(z) - C_{m-1}(z).$$

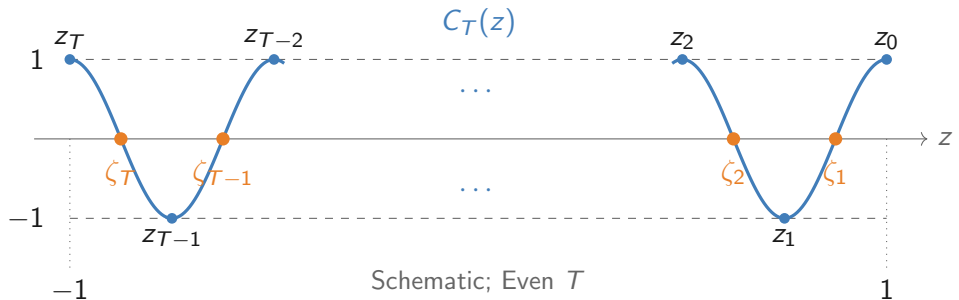
$\deg(C_T) = T$.

Exactly $T+1$ Alternating Extrema

$$z_j = \cos\frac{j\pi}{T}, \qquad C_T(z_j) = (-1)^j, \qquad j = 0, \ldots, T.$$

Chebyshev Roots Between Consecutive Extrema

$$\zeta_j = \cos\left(\frac{(2j-1)\pi}{2T}\right), \qquad C_T(\zeta_j) = 0, \qquad j = 1, \ldots, T.$$



$$z_j < \zeta_j < z_{j-1}, \qquad z_j = \cos\frac{j\pi}{T}, \quad j = 1, \ldots, T.$$

Chebyshev Optimality: Proof By Contradiction

$\deg(s) \le T, \quad \|s\|_\infty \le 1$
Suppose $s(c) > C_T(c) > 0$ (positivity: recurrence identity).

$$\alpha = \frac{C_T(c)}{s(c)} \in (0,1), \qquad h(z) = C_T(z) - \alpha s(z).$$

Interior Zeros

$$h(z_j) = (-1)^j - \alpha s(z_j), \qquad |\alpha s(z_j)| < 1.$$

Thus $h(z_j)$ has sign $(-1)^j$: one zero in each (z_{j+1}, z_j),
$j = 0, \ldots, T-1$, by continuity. The T intervals are disjoint.
Outside Zero

$$h(c) = C_T(c) - \alpha s(c) = 0, \qquad c > 1.$$

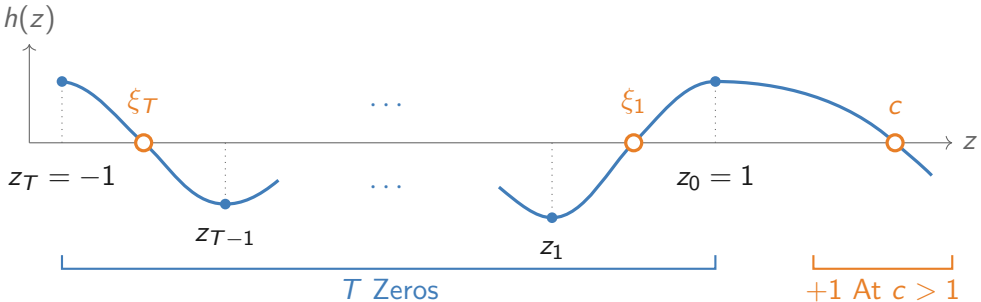
So $h \not\equiv 0$ has $T+1$ distinct zeros, but $\deg(h) \le T$: impossible.

$$\boxed{M_T = C_T(c).} \qquad \text{Attained by } s = C_T.$$

Chebyshev Optimality: Counting Polynomial Zeros

$$h = C_T - \alpha s, \qquad \deg(h) \le T, \qquad \operatorname{sign} h(z_j) = (-1)^j, \qquad h(c) = 0.$$

Hypothetical h (Even T)



$$\xi_j \in (z_j, z_{j-1}), \quad h(\xi_j) = 0 \ (j = 1, \ldots, T), \qquad \boxed{T + 1 \text{ distinct zeros} > \deg(h).}$$

Normalizing The Optimal Spectral Polynomial

Normalization At c

$$r^\star(z) = \frac{C_T(z)}{C_T(c)}, \qquad r^\star(c) = 1, \qquad \|r^\star\|_\infty = \frac{1}{C_T(c)}.$$

Spectral Variable λ

$$\boxed{p_T^\star(\lambda) = \frac{C_T\left(\dfrac{L+\mu-2\lambda}{L-\mu}\right)}{C_T\left(\dfrac{L+\mu}{L-\mu}\right)}.}$$

$$p_T^\star(0) = 1, \qquad \max_{\lambda\in[\mu,L]} |p_T^\star(\lambda)| = \frac{1}{C_T(c)} = \rho_T^\star.$$

From Chebyshev Roots To GD Stepsizes

Roots Of C_T: Cosine Identity

$$\zeta_j = \cos\left(\frac{(2j-1)\pi}{2T}\right), \qquad j = 1, \dots, T.$$

Roots Of $p_T^\star$

$$\gamma_j = \frac{L+\mu}{2} - \frac{L-\mu}{2}\cos\left(\frac{(2j-1)\pi}{2T}\right) \in (\mu, L).$$

Same Roots; $p_T^\star(0) = 1$

$$p_T^\star(\lambda) = \prod_{j=1}^{T}\left(1 - \frac{\lambda}{\gamma_j}\right), \qquad \boxed{\eta_{j-1} = \frac{1}{\gamma_j} > 0.}$$

Bounding The Optimal Polynomial Amplitude

$$q = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} \in (0,1), \qquad c = \frac{L+\mu}{L-\mu} = \frac{1}{2}(q+q^{-1}).$$

Denominator: The Same Recurrence

Set $u_m = (q^{-m}+q^m)/2$. It satisfies

$$u_0 = 1, \qquad u_1 = c, \qquad u_{m+1} = 2cu_m - u_{m-1}.$$

The initial values and recurrence agree with $C_m(c)$, so

$$C_T(c) = \tfrac{1}{2}(q^{-T}+q^T) > 0.$$

Numerator: $|C_T(z)| \le 1$ On $[-1,1]$

$$\boxed{\max_{\lambda\in[\mu,L]} |p_T^\star(\lambda)| = \frac{1}{C_T(c)} = \frac{2q^T}{1+q^{2T}} \le 2q^T.}$$

Chebyshev Steps: Distance Rate And Complexity

Symmetry gives $Q = V\Lambda V^\top$, $V^\top V = I$, $\lambda_i \in [\mu, L]$. Hence

$$\begin{aligned}\|e_T\| &= \|p_T^\star(Q)e_0\| \le \|p_T^\star(Q)\|_2 \, \|e_0\| \\ &= \max_i |p_T^\star(\lambda_i)| \, \|e_0\| \le 2q^T \|e_0\|.\end{aligned}$$

Iteration Complexity $s = 1/\sqrt{\kappa} \in (0,1)$

$$\log(1/q) = \log \frac{1+s}{1-s} = \int_0^s \frac{2}{1-u^2}\, du \ge 2s \quad \Longrightarrow \quad q^T \le e^{-2T/\sqrt{\kappa}}.$$

For relative error $0 < \epsilon \le 1/2$, it suffices to take

$$T \ge \frac{\sqrt{\kappa}}{2} \log \frac{2}{\epsilon}, \qquad T = O\big(\sqrt{\kappa}\log(1/\epsilon)\big)\,.$$

From Quadratic To General Convex Objectives

Chebyshev

Fixed quadratic Hessian Q.

Known spectral bounds $0 < \mu < L$.

Silver

General convex, L-smooth f.

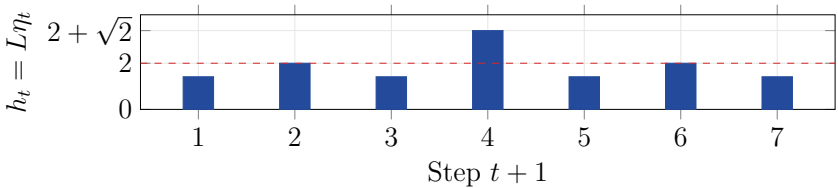
A minimizer exists.

Recursive Construction Of The Silver Schedule

$h_t = L\eta_t, \qquad \rho = 1 + \sqrt{2}, \qquad |H_k| = 2^k - 1$

$$\boxed{H_1 = [\sqrt{2}], \qquad H_{k+1} = [\, H_k,\ 1 + \rho^{k-1},\ H_k\,], \quad k \geq 1.}$$

$$H_1 = [\sqrt{2}],$$
$$H_2 = [\sqrt{2},\ 2,\ \sqrt{2}],$$
$$H_3 = [\sqrt{2},\ 2,\ \sqrt{2},\ 2+\sqrt{2},\ \sqrt{2},\ 2,\ \sqrt{2}].$$



Convergence Rate Of The Silver Schedule

Setting And Algorithm

$f : \mathbb{R}^d \to \mathbb{R}$ convex, L-smooth, $L > 0$; $x^\star \in \arg\min f$; $R = \|x_0 - x^\star\|$.

$$x_{t+1} = x_t - (h_t/L)\nabla f(x_t), \qquad (h_0, \dots, h_{T-1}) = H_k.$$

Theorem (Altschuler–Parrilo (2023), Theorem 1.1)

For every integer $k \geq 1$, at the block endpoint $T = 2^k - 1$,

$$f(x_T) - f^\star \leq \frac{LR^2}{1 + \sqrt{4\rho^{2k} - 3}}.$$

$$\rho^k = (T+1)^{p_{\mathrm{sil}}}, \qquad p_{\mathrm{sil}} = \log_2(1+\sqrt{2}) \approx 1.2716.$$

Altschuler–Parrilo, Theorem 1.1 and Definition 2.1.

Worst-Case Error Of Predetermined Schedules

Algorithm Class Fix $L > 0$ and an integer horizon T.

$$x_{t+1} = x_t - \eta_t \nabla f(x_t), \qquad \eta_0, \dots, \eta_{T-1} \geq 0.$$

Steps may depend on L, T, but are fixed before the objective and observed gradients.

Definition (Worst-Case Normalized Last-Iterate Error)

$$\mathcal{R}_T(\eta) = \sup_{\substack{d \geq 1,\ f:\mathbb{R}^d \to \mathbb{R} \text{ convex, } L\text{-smooth} \\ x^\star \in \arg\min f,\ x_0 \neq x^\star}} \frac{f(x_T) - f^\star}{(L/2)\|x_0 - x^\star\|^2}.$$

Ma–Chen Lower Bound For Predetermined Steps

Predetermined, Nonnegative GD Steps

Theorem (Ma–Chen (2026), Theorem 2.1)

Let $p_{\mathrm{MC}} = \sqrt{2+\sqrt{3}} \approx 1.9319$. For every $p \in (p_{\mathrm{MC}}, 2)$ there is $c_p > 0$ such that, for every $T \geq 1$ and every such schedule,

$$\mathcal{R}_T(\eta) \geq \frac{c_p}{(T+1)^p}.$$

Ma–Chen, A Lower Bound For Stepsize-Based Acceleration Of Gradient Descent.

Ye–Liu Lower Bound And The Silver Exponent

Theorem (Ye–Liu (2026), Theorem 1.1)

There is an absolute $C_0 > 0$ such that, for every sufficiently large T and every predetermined nonnegative schedule,

$$\mathcal{R}_T(\eta) \geq T^{-(p_{\mathrm{sil}}+\delta_T)}, \qquad \delta_T = C_0\sqrt{\frac{\log\log T}{\log T}} \longrightarrow 0.$$

Ye–Liu, Silver Rate Is (Almost) Optimal For Gradient Descent.

Summary Of Convergence Guarantees

Setting	Step Choice	Output And Certificate
Smooth, lower bounded	$1/L$	$\min_{t<T} \|\nabla f(x_t)\|^2 = O(1/T)$
Smooth convex	$1/L$	Last gap $\leq LR^2/(2T)$
Smooth strongly convex	$1/L$	Last gap $\leq (1-\mu/L)^T \Delta_0$
Subgradient (fixed)	$R/(G\sqrt{T})$	Averaged gap $\leq RG/\sqrt{T}$
Strongly convex quadratic	Chebyshev schedule	Distance at T: $D_T \leq 2R\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^T$
Smooth convex, silver	H_k/L	Last gap $O(LR^2/T^{1.2716\ldots})$

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Altschuler–Parrilo: Silver Stepsize Schedule For Smooth Convex Optimization.

Ma–Chen: A Lower Bound For Stepsize-Based Acceleration Of Gradient Descent.

Ye–Liu: Silver Rate Is (Almost) Optimal For Gradient Descent.